

Amar Merabti

AI Engineer & Data Scientist — CIFRE PhD Candidate

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Profile

M2 student in AI & Data Science (DCI) at Université Paris Cité, ranked 1st in my class (16/20). Research-oriented profile in deep learning and NLP: from implementing Transformer architectures from scratch to designing complete experimental protocols (benchmark, ablation, paired tests). Applying for a CIFRE (industrial) PhD starting October 2026.

Technical Skills

Machine Learning: Scikit-Learn, linear and logistic regression, Ridge/Lasso, decision trees, Random Forest, Gradient Boosting (XGBoost, LightGBM), SVM, k-NN, Naive Bayes, MLP, K-means, DBSCAN, hierarchical clustering, PCA, t-SNE, Isolation Forest
Deep Learning: PyTorch, TensorFlow, Keras, multi-GPU, tabular foundation models (TabPFN, TabSTAR)
NLP & LLM: Transformers, BERT/DeBERTa, NLI, BGE-M3 embeddings, GPT-2, LLaMA, Mistral, DeepSeek, fine-tuning, RAG
Data Engineering: Kafka, Spark, PySpark, MongoDB, SQL/NoSQL, Pandas, NumPy
Data Quality: faithfulness verification
Time Series: STL, LSTM, Transformers, imputation
Evaluation: ablation, paired tests, calibration
Programming: Python, Java, C/C++
Web & BI: Power BI, SSIS, HTML/CSS/JS

Professional Experience

Data Science Intern — Nricher, Paris

Feb. – Aug. 2026

- Production product-catalogue enrichment chain: rule- and lexicon-based attribute extraction algorithms, normalisation to the client reference schema, analysis of the resulting check files.
- Master's thesis**, supervised by Pr. Salima Benbernou — verification of extracted values against the source text alone, the client catalogue being unreliable: verbalisation of each attribute/value pair into a hypothesis, splitting of the description into premises and aggregation of natural language inference scores (mDeBERTa-v3). Benchmark over 4 production catalogues: 35 attributes, 30,919 rows, ground truth without manual annotation (source agreement, controlled error injection), 5,635 paired runs.
- Comparison of 3 architectures: learning-free NLI rule (0.808 balanced accuracy), TabPFN-3 on NLI scores + PCA-reduced BGE-M3 embeddings (0.925), TabSTAR reading the text natively (0.898, $\times 28$ cost per run). Paired ablation (Wilcoxon): -8.9 points without the 4 inference scores against -1.9 without the 50 components — what decides is the orientation of the text representation, not its richness. Calibrated decision: 68% of values automated at 99% precision.

AI Developer (Machine Learning) — ENAGEO (Sonatrach group), Boumerdès

July 2023 – July 2024

- Part-time fixed-term contract alongside the M1 SII: predicting well measurements (elasticity, density...) that are missing or costly to acquire, treating each depth-indexed series as a time series.
- Within-well imputation with LSTM and RNN models (trend, seasonality, cycles): MAE 12.7, RMSE 15.2 (LSTM).
- Cross-well generalization (LSTM), vs. the internship: correlation $0.52 \rightarrow 0.82$, MAE $21.4 \rightarrow 15.8$, RMSE $27.6 \rightarrow 20.2$.

Machine Learning Intern — ENAGEO (Sonatrach group), Boumerdès

Feb. – June 2023

- Bachelor's thesis** (June 2023) — prediction of geomechanical properties of reservoir rock from the well logs of 3 boreholes, under cross-well generalization (testing on an unseen well). Linear regression ($r = 0.15$), RBF-SVM (0.29), 100/50 ReLU MLP (0.52, MAE 21.4): correlation too low on heterogeneous formations. Delivered a Java Swing / Python application (MySQL, UML).

Research Projects

Kabyle LLM from Scratch

Personal research project · 2026

Design and from-scratch training of a generative language model (36.3M parameters) for Kabyle, a low-resource language. Llama-style decoder-only Transformer in PyTorch (10 layers, 8 heads, $d_{model} = 512$, context 512, RMSNorm, RoPE, SwiGLU), custom BPE tokenizer. Pretraining on 107M tokens (one Tesla T4), then fine-tuning: bits per character reduced by 3.3%.

Open-Source LLM Benchmark

Research project, TER (M1) · 2025

Empirical comparative evaluation of three 7B LLMs (LLaMA 2, Mistral, DeepSeek-R1-Distill-Qwen) on GSM8K (8-shot), HellaSwag (0-shot), and HumanEval (0-shot, sandboxed execution), on $4 \times$ NVIDIA L4 via Ollama. Reproducible protocol, analysis of Python error types and of the accuracy/latency trade-off — DeepSeek 78.0% on HumanEval, Mistral 61.2% on HellaSwag.

Temporal Expression Extraction with Transformers

Research project (M2) · 2025

Cross-domain evaluation of a BERT token-classification model (Lange et al., 2021) for TIMEX3 typing, across 3 corpora with contrasting registers (historical, clinical, tweets). Automated pipeline and gold-standard alignment — F1 up to 95.5%.

Time Series Imputation

M1, supervised by Pr. Themis Palpanas · 2025

Comparison of 6 imputation strategies on 3,127 electricity consumption series of 12,864 points, in PyTorch (GPU-accelerated STL, LSTM autoencoder, Partial Convolution U-Net, self-supervised Transformer) — best MAE by GPU-STL (0.262).

Academic Projects

Data Quality — Inconsistency Detection & Correction

M2 · 2025

Multi-criteria pipeline (completeness, accuracy, consistency, uniqueness): regression-based imputation ($r \approx 0.78$ between fare and distance), geometric reconstruction of GPS coordinates (Haversine), correction of temporal inconsistencies.

Real-Time E-Commerce Recommendation Engine

M2 · 2025

Distributed streaming architecture (Kafka $\rightarrow$ PySpark $\rightarrow$ MongoDB $\rightarrow$ Streamlit): catalog of 24,255 products derived from 885K events, hybrid recommendation and online relevance scoring.

Education

2024–2026 **M1 & M2 Data, Knowledge and Intelligence (DCI), Ranked 1st in class** — Université Paris Cité

2020–2024 **B.Sc. Computer Science + M1 SII** — USTHB, Algiers

2020 **Baccalaureate, Experimental Sciences, with Honors** — Algeria

Languages & Interests

Languages: French (bilingual) · English (bilingual) · Arabic (fluent) · Kabyle (native)

Interests: AI & Sciences · Languages & Linguistics · Football · Music · Photography